

Designing smart contracts with blockchain data to reveal quality information in an agricultural supply chain with multiple suppliers

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2025.11.29

福州, 第八届智能制造系统工程学术会议

Evidence about misalignment of incentive（激励错位现象的证据）

- *"We find that, in a competitive setting, blockchain adoption does not arise in equilibrium even when that outcome enhances the total welfare of all the participants. This is because the participants who stand to gain from blockchain adoption are distinct from those who control whether the blockchain is adopted"* (Iyengar et al., 2023).



- 胡雯 et al., (2024)系统地梳理了砀山酥梨、潍坊韭菜、西湖龙井茶三个实践案例，研究发现目前农产品区块链溯源技术应用仍面临溯源成本高、产品溢价难预期、利益分配机制不健全等关键挑战。

[1] Iyengar, G., Saleh, F., Sethuraman, J., & Wang, W. (2023). Economics of Permissioned Blockchain Adoption. *Management Science*, 69(6), 3415–3436.

<https://doi.org/10.1287/mnsc.2022.4532>

[2] 胡雯, 黄季焜, 陈富桥, 饶思航, & 易红梅. (2024). 基于区块链技术的农产品质量安全追溯体系: 实践、挑战与建议. *农业经济问题*, 05.

Research questions

1. Hidden actions $\Rightarrow$ Uncertain quality of post harvest;
2. Misalignment of incentivizing blockchain adoption

Our study answers three research questions:

- How to design a menu of smart contracts that simultaneously reveals the farmers' hidden actions and encourages voluntary blockchain adoption, regardless of the product's verifiable features;
- How to transform the theoretical form of optimal contracts into their practical form by applying machine learning techniques with blockchain data;
- How to reduce the misalignment of incentives among farmers if the retailer manages blockchain adoption.

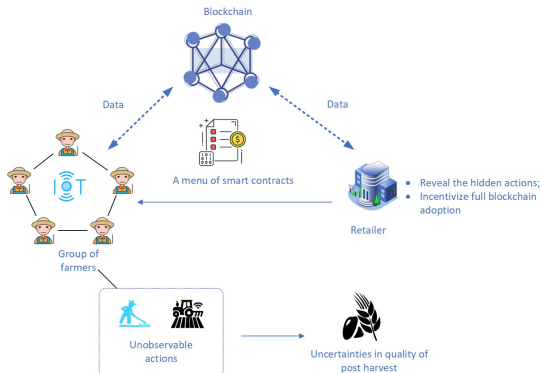
Literature Review

Our study relates to three streams of literature, including the incentive for blockchain adoption, the proactive disclosure of quality information, and the reactive revelation of quality information.

- **Permissioned blockchain adoption incentive mechanism**
Alharbi & Hussain(2025),Parra-López et al.(2025),Iyengar et al.(2023) et al
- **Proactively disclosing information on product quality**
Guan & Chen(2017),Shi et al.(2023),Liu et al.(2024) et al
- **Revelation of private information on the quality of agricultural products**
Babich & Tang(2012),Gao et al.(2020),Qian & Olsen(2022) et al

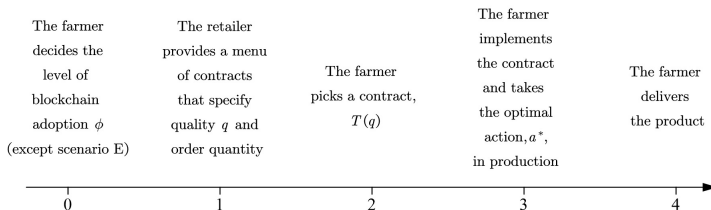
Contributions

- We incorporate the Bayesian learning technique into the theoretical model that is based on incentive theory.
- Our model is a dynamic model in which the downstream player's transfer payment differs with the random quality level.
- We jointly consider three types of uncertainties: a farmer's prior belief about the realized product quality, the consumers' preference for a specific quality level, and the random product quality.



Problem description and model settings

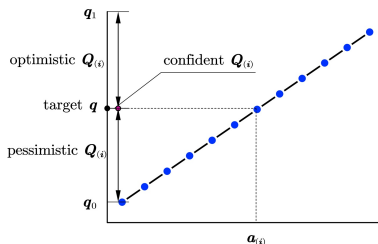
- We consider a supply chain consisting of a retailer and multiple farmers.
- The retailer is the leader, while farmers are the followers.
- The retailer determines the transfer payment $T(q)$ and retail price $p(q)$, according to the farmer's reported quality level.
- Farmers determine their optimal action a in production.
- Let $\phi \in [0, 1]$ denote the blockchain adoption level.



Model settings: the type of farmers

We consider three types of farmers who may have different risk attitudes regarding the estimation of the realized quality level, $Q_{(i)}$.

$$\mathbb{E}^{Prior} \left[Q_{(i)} \mid \text{target } z \right] \stackrel{\text{def}}{=} \begin{cases} z & \text{if } i = \text{confident farmer (CF)} \\ \int_z^{q_1} \tau \, dF(\tau) & \text{if } i = \text{optimistic farmer (OF)} \\ \int_{q_0}^z \tau \, dF(\tau) & \text{if } i = \text{pessimistic farmer (PF)} \end{cases}$$



♣ For instance, a farmer's targeted quality is q before production, and takes action a during production. After production, the realized quality is $Q_{(i)}$.

The basic model: lump sum contract(Scenario L)

Each farmer of various types seeks the optimal action that maximizes their expected profit

$$\max_a \pi_s = T - \underbrace{(1 - \phi) m \sigma^2 \mathbb{E}^{Priori} [Q_{(i)}^2 | target \ q]}_{\substack{\text{potential losses} \\ - \frac{1}{2} \beta a^2 - \frac{1}{2} k \phi^2}} - \underbrace{c \mathbb{E}_{\Theta} [D | q]}_{\text{normal production expense}}$$

It yields

$$\begin{cases} \dot{T} = \beta a + c \theta + (1 - \phi) m \sigma^2 \frac{\partial \mathbb{E}^{Priori} [Q_{(i)}^2 | target \ q]}{\partial q} - bcu \\ \dot{p} = u \end{cases}$$

The expected profit of the retailer is as follows:

$$\max_{T, p, u} \mathbb{E}_{q \sim F} [\pi_R] = \int_{q_0}^{q_1} [p(q) \mathbb{E}_{\Theta} [D | q] - T(q)] dF(q)$$

The basic model: lump sum contract(Scenario L)

Theorem 2

The optimal blockchain adoption level depends on the farmer type. The general form of the optimal blockchain adoption is as follows:

$$\phi^{L(i)*} = \frac{m\sigma^2}{k} \mathbb{E}^{Priori} \left[Q_{(i)}^2 | target \ q_0 \right] \quad (10)$$

if $m\sigma^2 \mathbb{E}^{Priori} \left[Q_{(i)}^2 | target \ q_0 \right] \geq k$, then $\phi^{L*} = 1$. The following statements hold true:

- (i) For a confident farmer, $\phi^{LCF*} = \frac{m\sigma^2 q_0^2}{k}$;
- (ii) For an optimistic farmer, $\phi^{LOF*} = \frac{m\sigma^2}{k} \int_{q_0}^{q_1} \tau^2 dF(\tau)$;
- (iii) For a pessimistic farmer, $\phi^{LPF*} = 0$;
- (iv) $\phi^{LOF*} > \phi^{LCF*} > \phi^{LPF*}$

The lump-sum contract is ineffective in ensuring full blockchain adoption among all farmer types. The theorem explains why failures in adopting blockchain systems can occur in practice.

Revenue sharing scheme (Scenario R)

The retailer transfers a proportion, $t(\phi)$, of their revenue to the farmer. Each farmer of various types seeks the optimal action that maximizes their expected profit:

$$\max_a \pi_s = \underbrace{T + t(\phi) p \mathbb{E}_{\Theta} [D | q]}_{\text{revenue sharing package}} - (1 - \phi) m \sigma^2 \mathbb{E}^{Priori} \left[Q_{(i)}^2 | \text{target } q \right] - c \mathbb{E}_{\Theta} [D | q] - \frac{1}{2} \beta a^2 - \frac{1}{2} k \phi^2$$

The expected profit of the retailer is as follows:

$$\max \pi_r = \int_{q_0}^{q_1} [(1 - t(\phi)) p \mathbb{E}_{\Theta} [D | q] - T] dF(q)$$

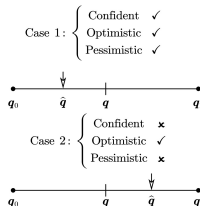
Stepwise contract for verifiable quality (Scenario S)

If the realized quality exceeds the threshold level, the farmer can claim the bonus of the amount of $B(\phi)(q - \hat{q})^+$ after production.

$$\max_a \pi_s = \left\{ \underbrace{T(q) + B(\phi)(Q_{(i)} - \hat{q})^+}_{\text{stepwise package}} - (1 - \phi) m \sigma^2 \mathbb{E}^{\text{Priori}} [Q_{(i)}^2 | \text{target } q] - c \mathbb{E}_{\Theta} [D | q] - 0.5 \beta a^2 - 0.5 k \phi^2 \right\} \quad (16)$$

Note that the stepwise contractual scheme consists of two parts: contract $T(q)$ and bonus $B(\phi)(Q_{(i)} - \hat{q})^+$. The retailer's profit function is as follows:

$$\max_{p, T} \mathbb{E}[\pi_r] = \int_{\hat{q}}^{q_1} \{p \mathbb{E}_{\Theta} [D | q] - T - B(\phi)(q - \hat{q})\} dF(q) \quad (17)$$



Design of data-enabled contracts (Scenario D)

The interdependency between the realized quality Q and the input data $\mathbf{x}$ is captured by the following equation:

$$Q = \mathbf{w}^T \mathbf{x} + \epsilon, \epsilon \sim \mathcal{N}(0, \zeta^2)$$

Our goal is to estimate the coefficient vector through learning from the input-output data pair, $data = \{Q_1, \mathbf{x}_1\}, \{Q_2, \mathbf{x}_2\}, \dots, \{Q_M, \mathbf{x}_M\}$. The problem is to maximize the log function of the posterior probability as indicated in Eq.(26).

$$\max_{\mathbf{w} \in \mathbb{R}^n} \ln p(\mathbf{w} | \mathbf{X}, \mathbf{Q}) \Leftrightarrow \min_{\mathbf{w} \in \mathbb{R}^n} \left\{ \frac{1}{2\zeta^2} \|\mathbf{Q} - \mathbf{X}\mathbf{w}\|_2^2 + \frac{1}{2s^2} \|\mathbf{w}\|_2^2 \right\} \quad (26)$$

The coefficient vector $\mathbf{w}$ is estimated using the dataset, thereby obtaining the maximum a posteriori (MAP):

$$\hat{\mathbf{w}}_{MAP} = \left(\frac{1}{\zeta^2} \mathbf{X}^T \mathbf{X} + \frac{\mathbf{I}}{s^2} \right)^{-1} \frac{1}{\zeta^2} \mathbf{X}^T \mathbf{Q}$$

$\mathbf{I}$ is the n identity matrix.

Theorem 9

In the case of the farmers' full blockchain adoption the optimal data-enabled contract for all types of farmers is as follows:

$$T^{D*} = \left(\beta \hat{\mathbf{w}}_{MAP}^T \mathbf{x} + \frac{1}{2} c\theta + 2m\sigma^2 \hat{\mathbf{w}}_{MAP}^T \mathbf{x} \right) (q - q_0) + T_0$$

Roadmap of algorithms for choosing contracts

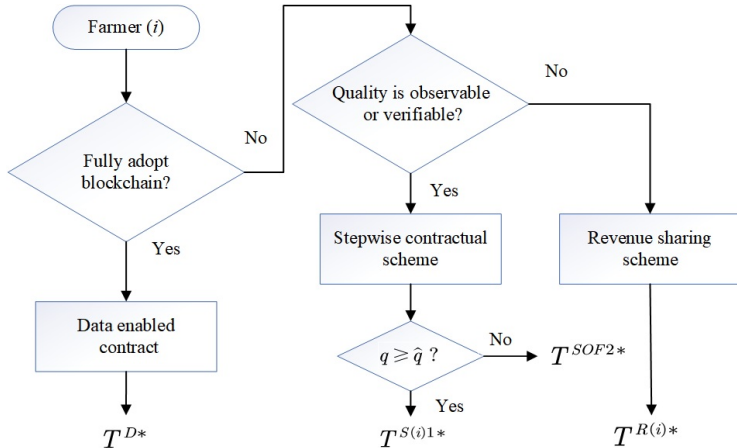
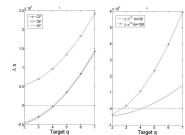
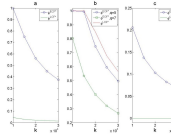
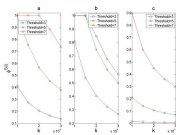
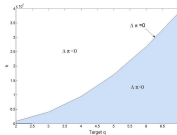
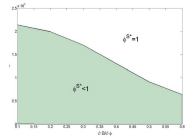
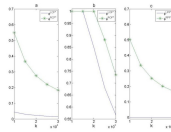
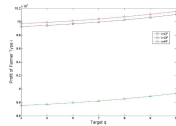
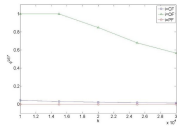


Fig 4.Roadmap of algorithms for choosing contracts

Numerical analysis



Conclusions and managerial implications

Research findings are:

- If the realized quality cannot be verified, the proposed revenue-sharing scheme can reveal the farmers' hidden actions, regardless of the farmer type.
- If the realized quality can be verified, the proposed stepwise scheme can simultaneously encourage full blockchain adoption, reveal the farmers' hidden actions, and adjust the minimum desired level of the realized quality.
- Full blockchain adoption does not necessarily occur whenever the retailer assumes control of the blockchain.

Managerial significance includes:

- Design smart contracts in agri-food supply chains.
- Our study findings align with those in the literature, such as Iyengar et al. (2023). A misalignment arises when supply chain members bearing blockchain's costs cannot derive a sufficient amount of benefit from using the blockchain.

